

SHALIN BHATTARAI

— Game Developer

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SUMMARY

Game developer with Unity/C# gameplay programming experience and a from-scratch JavaScript game engine: real-time physics, particle systems, animation state machines, and dynamic cameras. Shipped a browser fighting game to a live domain; contributing to a faculty-led Unity research simulation.

SKILLS

Game Development: Unity, C#, Unreal Engine 5, Gameplay Programming, Real-Time Physics, Particle Systems, Shader Graph, Animation State Machines, Collision Detection, Game Loops, Level Design, Input Handling

Programming: C#, C++, JavaScript (ES6+), Java, Python, SQL

Tools & Pipeline: Git, GitHub, Linux, Bash, Figma, HTML5/CSS3, Node.js, Agile Team Collaboration

PROJECTS

STORM (Simulated Tornado Module) – Interactive Simulation — Unity, C# 2026

- Contributing to a faculty-led research project (Dr. Gayathri Sadanala) by building an interactive, educational tornado simulation in Unity with C#.
- Achieved a convincing real-time storm environment by implementing physics systems, particle effects, dynamic weather, and environmental interactions.
- Raised visual quality across the module by developing character animations, level designs, and custom visual effects with Shader Graph.
- Turned the simulation into a teaching tool by integrating in-engine data visualizations and interactive quizzes on tornado formation and safety.

Vortex Combat – 2D Fighting Game — JavaScript, HTML5, CSS3 2025

- Shipped a complete browser fighting game to a custom domain (vortexcombat.com) in a 2-week sprint on a 2-person team, writing every engine system from scratch with zero external libraries.
- Delivered responsive, readable combat—movement, jumping, guarding, quick/strong attacks, stamina management, and character-specific specials such as a teleport—by engineering frame-based combat logic and animation state machines driving 30+ unique sprites per fighter.
- Built the core engine loop with collision detection, hit/hurt states, keyboard and gamepad input handling, and OOP architecture for 2 playable fighters and 4 hazard-filled stages (Temple, Palace, The Pit, Dead Pool).
- Amplified game feel by implementing a dynamic zoom-and-pan camera, particle effects, screen shake, and gameplay-synchronized audio.

J.A.R.V.I.S. – Real-Time Interactive AI Interface — JavaScript, Node.js, Web Speech API 2025

- Created a game-like HUD experience with boot sequences, hexagonal grid overlays, ripple animations, and live audio visualization by building a custom animation framework in vanilla JavaScript.
- Kept animations and voice interaction in sync at interactive frame rates by engineering asynchronous state management across the Gemini, ElevenLabs, and Web Speech APIs.

Additional Full-Stack Projects — PHP, MySQL, Flask, JavaScript 2024–2025

- Built !Nike, a full-stack e-commerce platform (PHP/MySQL: auth, cart, checkout), and an Instagram clone (Flask/MySQL: posts, likes, follows, messaging)—database and systems skills that carry into game backends.
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EXPERIENCE

Truman State University – eSports Program

Kirksville, MO

Web Developer

2025–Present

- Support the university's competitive gaming program by developing and maintaining its official website, publishing tournament information, event schedules, and team rosters for players and fans.
- Improved usability across devices by designing responsive layouts, customizing themes, and optimizing site performance in a live production environment.

Google Developer Student Club — Secretary

Aug 2024–May 2025

Namaste Nepal, Truman State University — Secretary

Jan 2024–Aug 2024

EDUCATION

Truman State University

Kirksville, MO

B.S. Computer and Information Sciences

Expected December 2026

Relevant Coursework: Data Structures & Algorithms, OOP (C++ & Java), Computer Architecture, Systems Programming, Internet Programming